

02464

Kunstig Intelligens og Menneskelig Kognition

Chapter 9 Concepts part I

Prototype and Exemplar based models

Defining concepts

- How do you define a concept?
 - Age old topic in philosophy
- Definition by crisp predicates
 - Dog
 - Hairy
 - Wags its tail
 - Four legs
 - Barks
 - Works?

Defining concepts

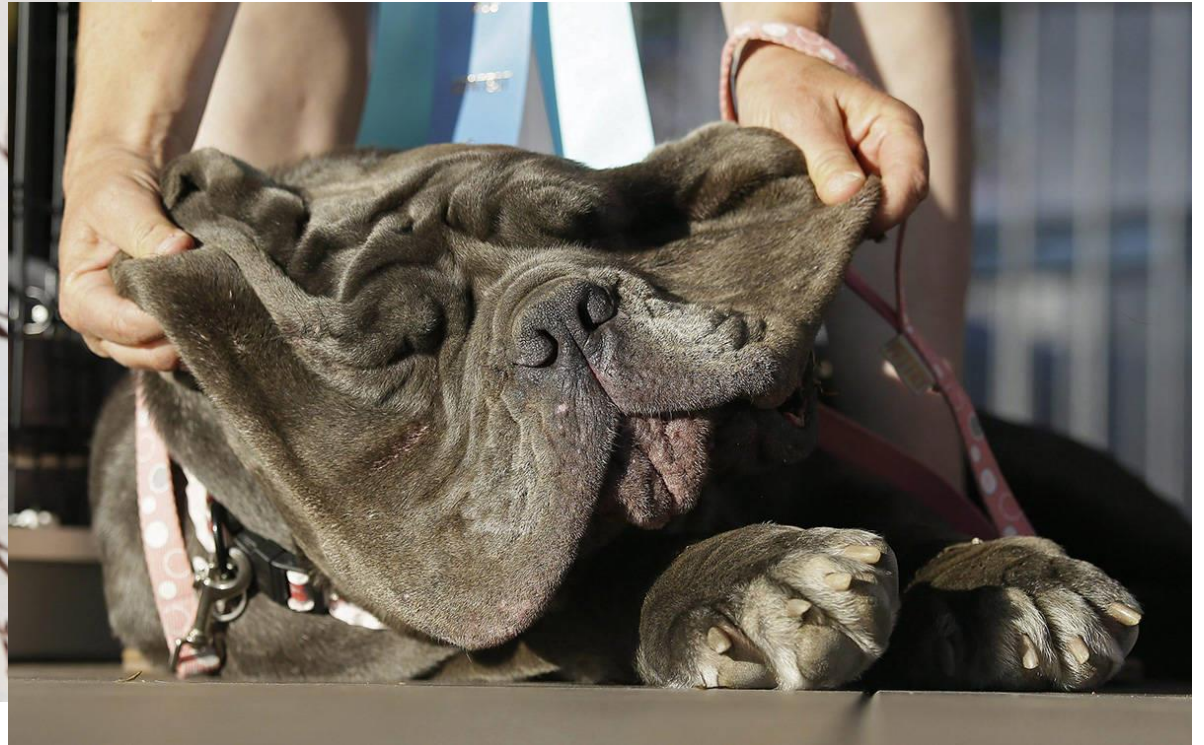
- How do you define a concept?
 - Age old topic in philosophy
- Definition by crisp predicates
 - Dog
 - Hairy Hairless Chihuahuas?
 - Wags its tail Couped Dobermann?
 - Four legs Three-legged disabled dog?
 - Barks Barkless Basenjis?
 - Harder than you think!
 - https://www.youtube.com/watch?v=Y1HVTNxwt7w&ab_channel=BBCScotland-Comedy
- Works even worse with complex concepts like love or world peace

Feature based similarity



Dog?

Dog?



Prototypes and Graded membership

- Some birds are more birdy than others
 - Some non-birds are more birdy than others
- As seen by *explicit* typicality rating tasks

TABLE 9.1

Fruit	Rating	Bird	Rating
Apple	6.25	Robin	6.89
Peach	5.81	Bluebird	6.42
Pear	5.25	Seagull	6.26
Grape	5.13	Swallow	6.16
Strawberry	5.00	Falcon	5.74
Lemon	4.86	Mockingbird	5.47
Blueberry	4.56	Starling	5.16
Watermelon	4.06	Owl	5.00
Raisin	3.75	Vulture	4.84
Fig	3.38	Sandpiper	4.47
Coconut	3.06	Chicken	3.95
Pomegranate	2.50	Flamingo	3.37
Avocado	2.38	Albatross	3.32
Pumpkin	2.31	Penguin	2.63
Olive	2.25	Bat	1.53

Exemplar based concepts

- If what I see before me resembles an apple I have seen *more than* anything else I have seen, then it is an apple
- Lassie resembles a dog I have seen
 - *And* does not resemble anything non-dog I have seen
- A chihuahua does not resemble a dog I have seen
 - *Or* resembles another type of animal *more*
- Graded membership but no prototype

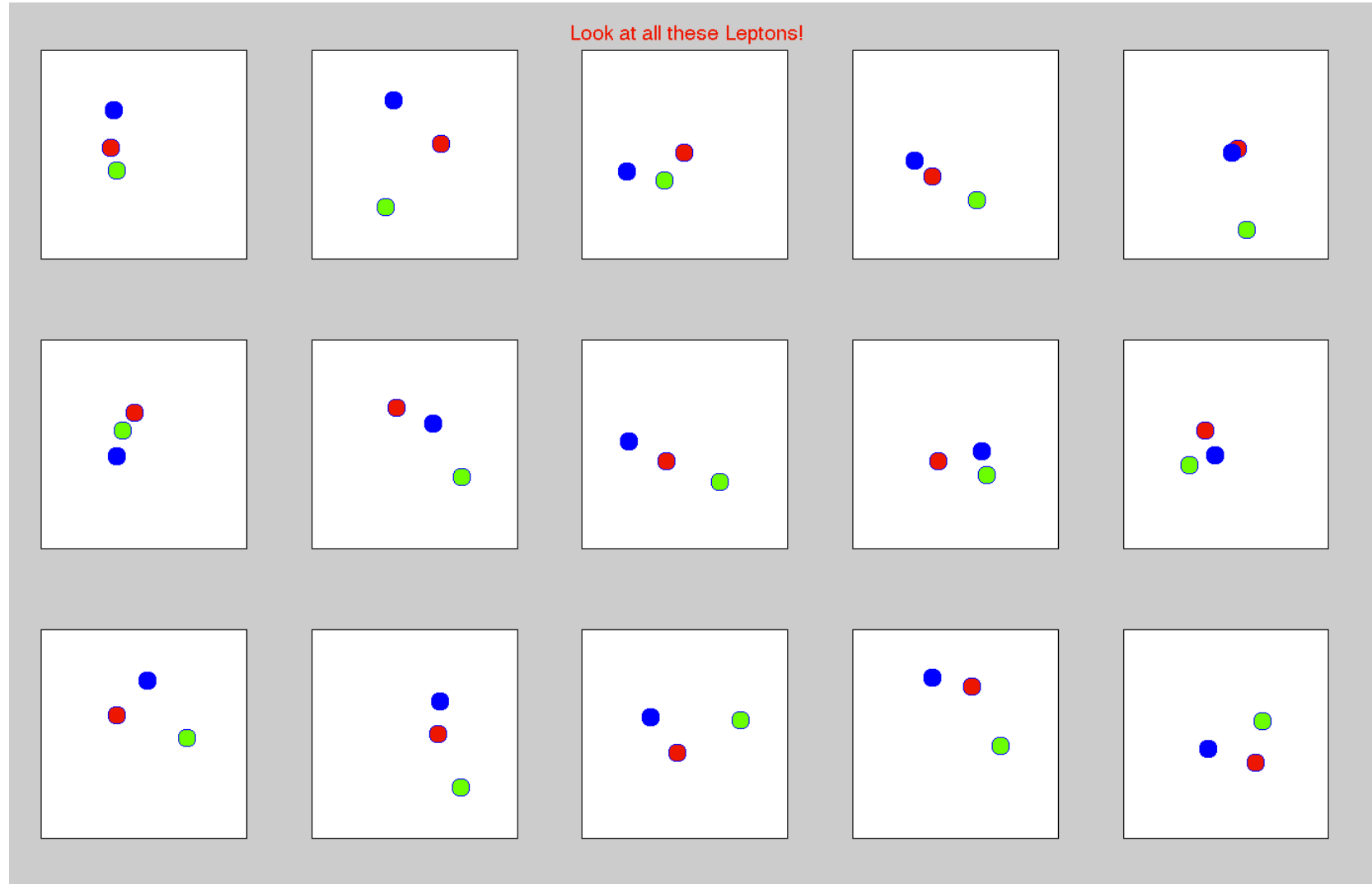
Prototypes vs. exemplars

- When can you form a prototype?
 - Children will correctly identify a dog given only a few examples
 - We might start out with an exemplar-based approach
- How many exemplars can you remember?
 - Is this an apple?
 - Wait a sec while I compare this to all the apples I have seen in my life

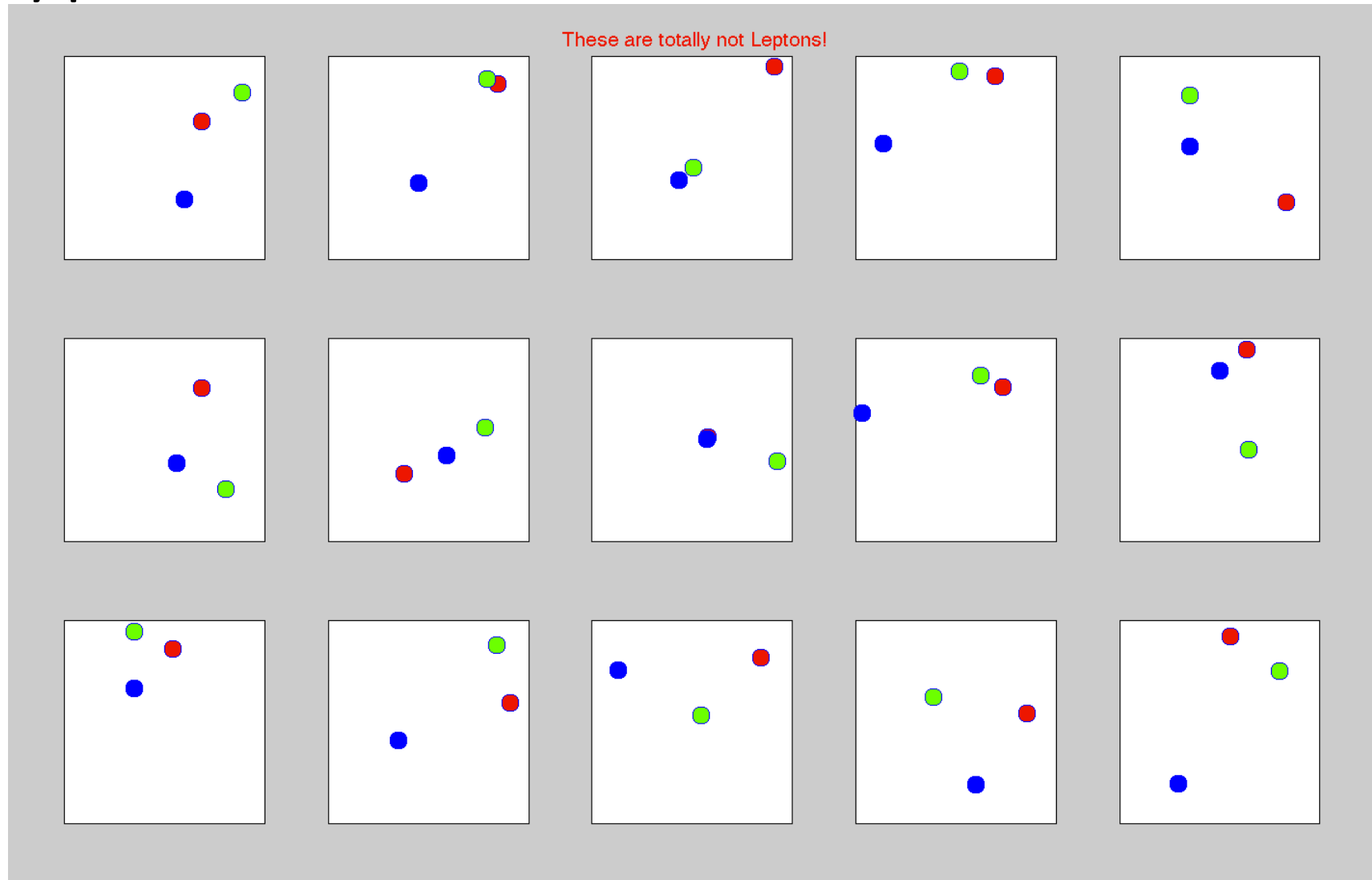
Prototypes vs. exemplars

- So..., which is it?
- Both
- Prototypes and exemplars are stored in our memory
 - Only a few exemplars can be stored
 - More recent
 - Special examples
- We use whichever one suitable for the task

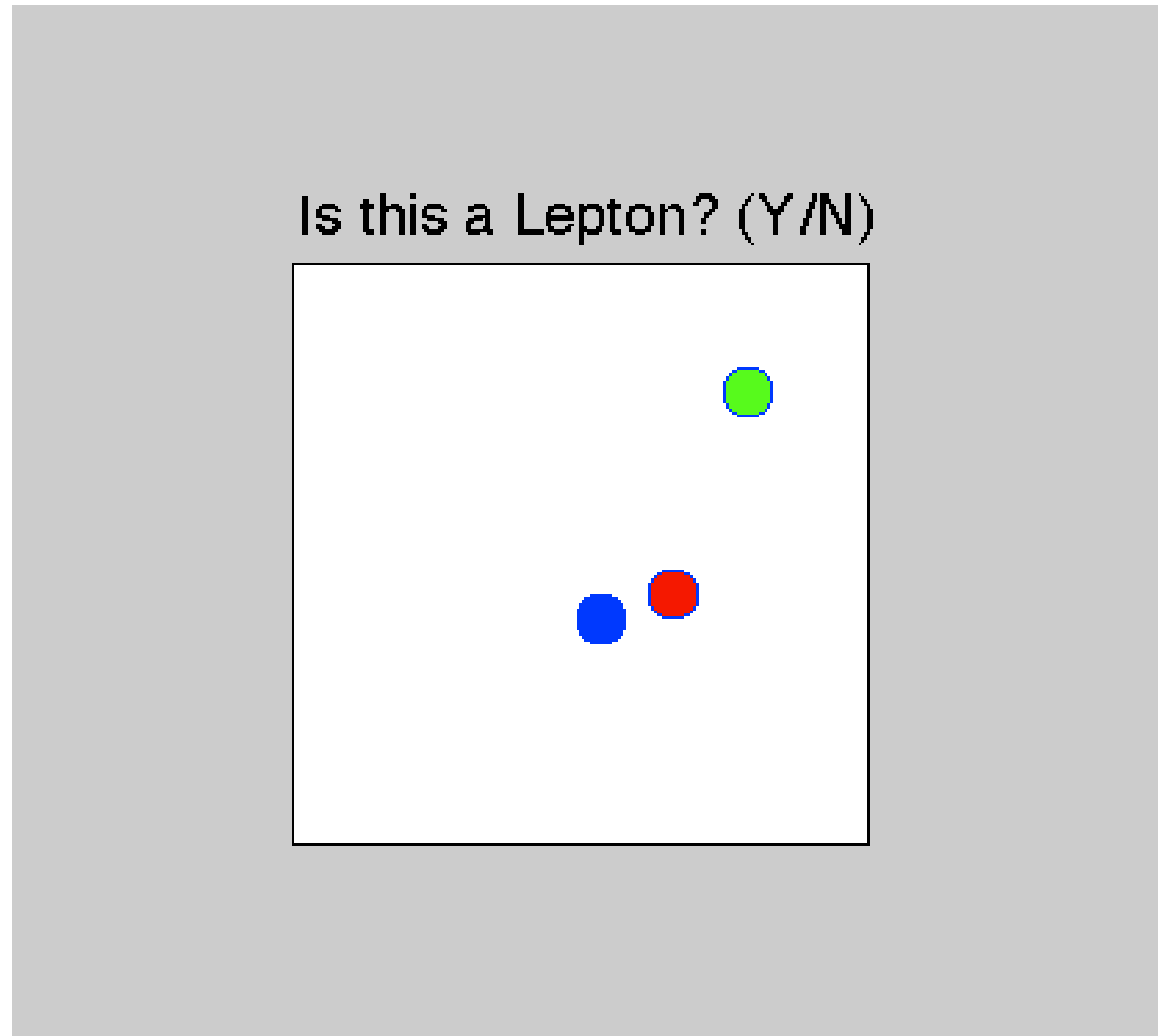
Prototype distortion task



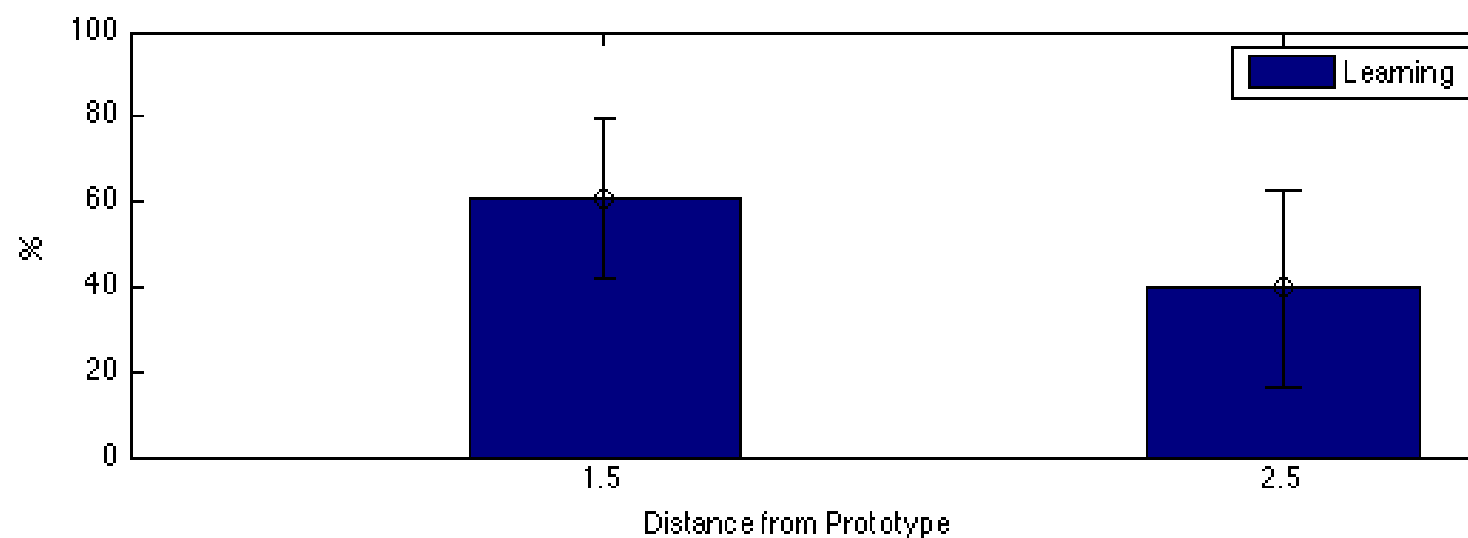
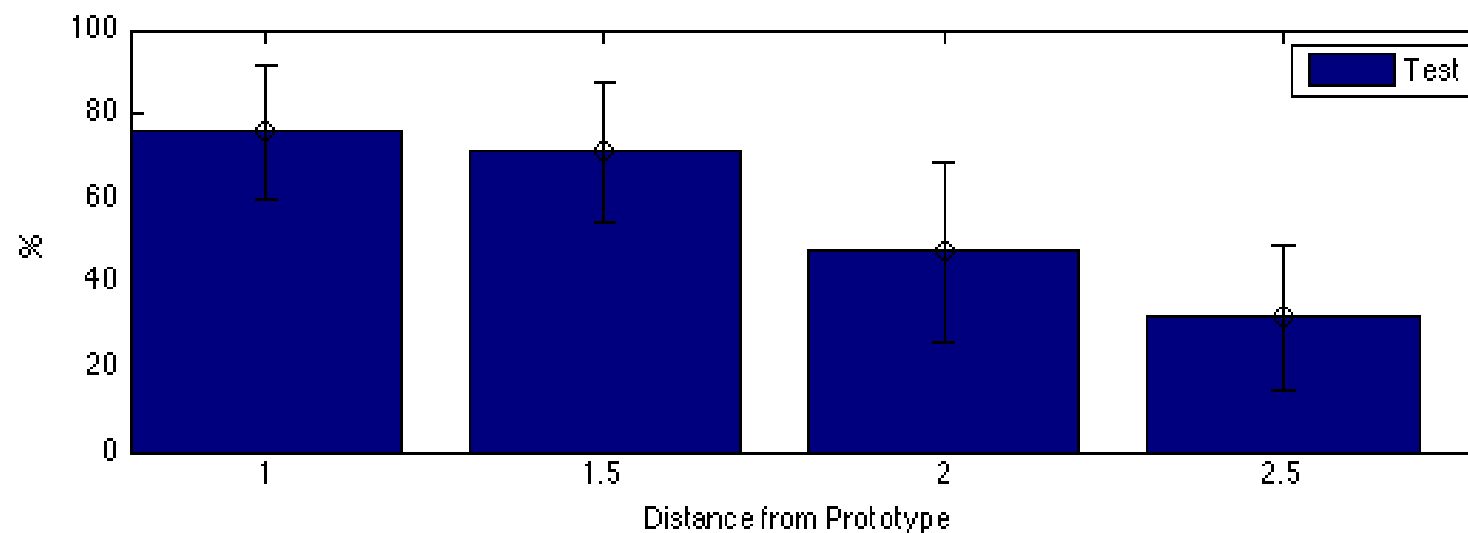
Prototype distortion task



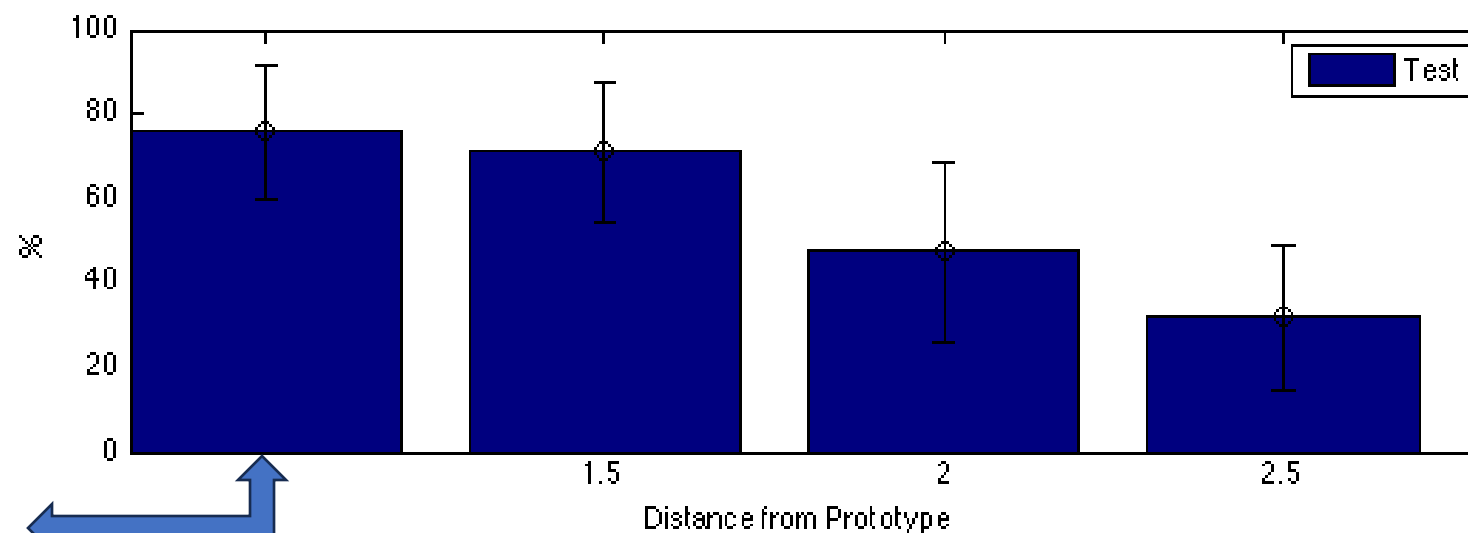
Prototype distortion task



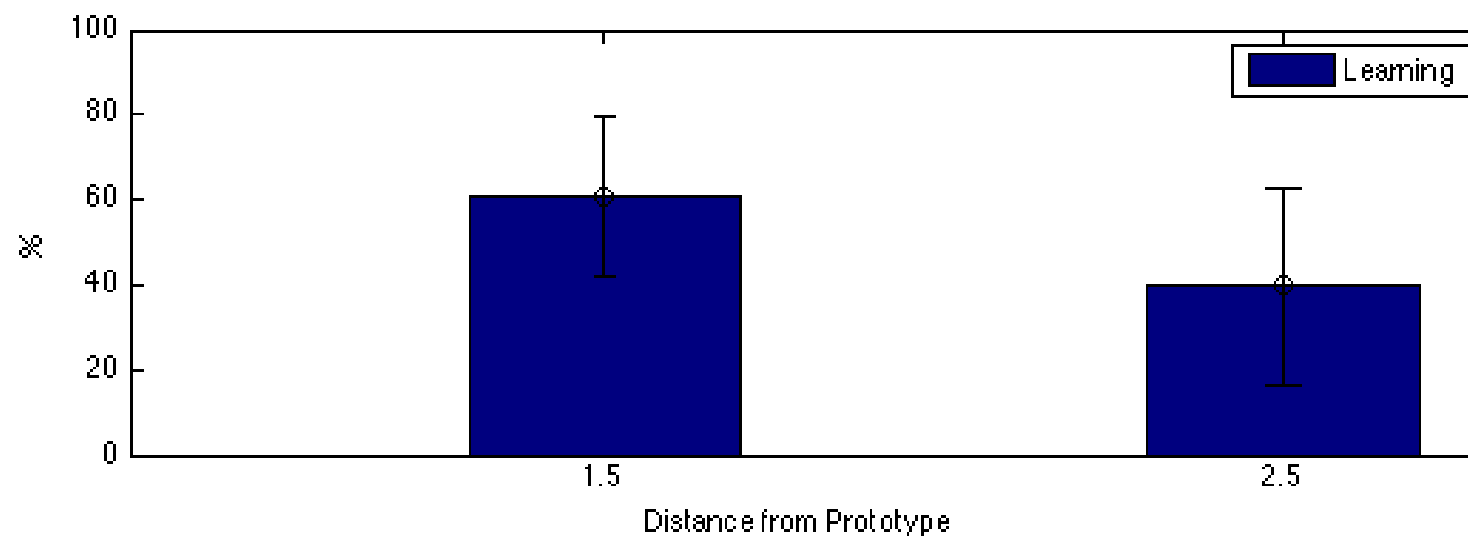
Results (17 subjects)



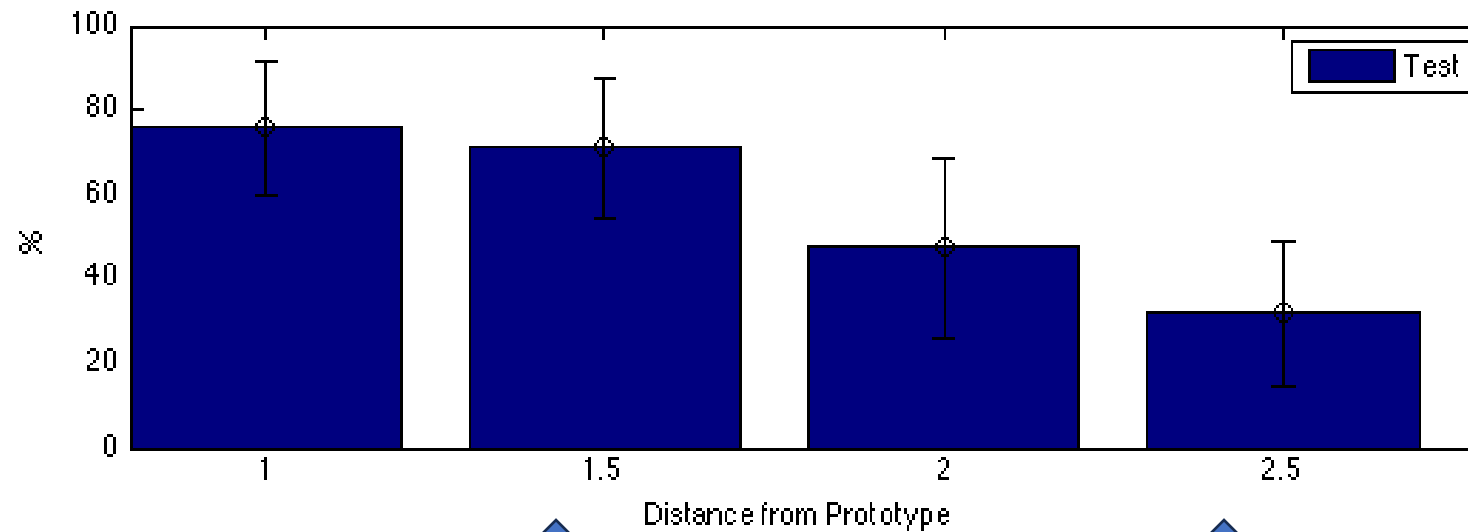
Results (17 subjects)



Closest to the prototype
But not to the exemplars



Results (17 subjects)



Prototype-based
observer sees no
difference

Exemplar-based
observer would be
perfect

